

Paul-Hieu V. Nguyen

Sept. 2026 paulh.nguyen@yahoo.com

EDUCATION	University of Wisconsin–Madison , Madison, WI Ph.D., Statistics, August 2022 - Present (Anticipated 05/27) Reed College , Portland, OR. B.A., Mathematics, May 2022 Thesis: “Inference for Random Forests”
EXPERIENCE	Global Decision Science Intern , American Express June 2026-Aug 2026 - Applied Bayesian Additive Regression Trees (BART) and XGBoost algorithms across classification and regression datasets (Gamer, New Accounts, Retention). - Employed Bayesian optimization for hyperparameter tuning. - Established gains in prediction for both overall performance and specific customer segments, compared to XGBoost production baseline. - Analyzed uncertainty quantification to evaluate model reliability and prediction uncertainty. Identified predictors associated with high uncertainty. - Presented results to senior leadership through powerpoint presentation and suggested ways to incorporate uncertainty quantification to reduce risk. Research Assistant , University of Wisconsin–Madison Aug 2022 - Present - Developed Bayesian Additive Regression Trees (BART) extensions in C++ and R. - Presented work at multiple machine learning and statistical conferences, published paper in machine learning journal, and built R package for model implementation. - Demonstrated improvements in predictive accuracy over current tree ensemble methods (Random Forest, XGBoost, BART) through cross-validation experiments across classification and regression datasets. - Awarded 1st Place at the CMU Sports Analytics Research Competition for development of Bayesian framework of the decathlon focused on uncertainty quantification and model interpretability. - Integrated large-scale real estate and energy datasets to model energy usage through surrogate models with uncertainty quantification. NSF REU Fellow , Pennsylvania State University May 2021 - Aug 2021 Research Assistant, Economics, Statistics , Reed College Jan 2020 - Aug 2020 Iowa Biostatistics Summer Fellow , University of Iowa May 2019 - Aug 2019
SKILLS	Programming: R, C++, Python, Latex, SQL. Software: Stan, Git Interests: Machine learning, Bayesian modeling, Uncertainty quantification, Tree ensemble methods
PUBLICATIONS	Nguyen, P-H.V. , Smoliga J.M., Lindaman B., Deshpande, S.K. (2026) “Quantifying the limits of human athletic performance: A Bayesian analysis of elite decathletes” <i>Journal of Quantitative Analysis in Sports</i> . Nguyen, P-H.V. , Yee R., Deshpande, S.K. (2025). “Oblique Bayesian additive regression trees”. <i>Transactions on Machine Learning Research</i> . Wojeik O.C., Olson S.D., Nguyen, P-H.V. , McConville K.S., Moisen G.G. and Frescino T.S. (2022). “GREGORY: A Modified Generalized Regression Estimator Approach to Estimating Forest Attributes in the Interior Western US.” <i>Frontiers in Forests and Global Change</i> .
HONORS & AWARDS	- 1st place, Carnegie Mellon Sports Analytics Research Competition, Oct 2025 - Advanced Opportunity Fellowship, University of Wisconsin–Madison, Aug 2022 - 2nd place, Undergraduate Research Project Competition (USPROC), June 2020 - 2020 Economics Summer Research Mintz Award, Reed College, Aug 2020 - Commendation for Excellence, Reed College, Aug 2019, 2020, 2021, 2022
SERVICE	Statistics Graduate Student Association, University of Wisconsin–Madison, 2023-Current